COMP-1807- Agile Development with SCRUM

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Group Report (Section 1)

Scrum Process

1.1 Product Vision

For: Customers (16-65 years old)
Who: Want a better experience
Product name: FastDeal
Is a: Mobile app for recent operating system (Android 10)
That: Allows customers to purchase mobile applications and subscriptions
Unlike: The present internet site
Our product: The services enable customers to purchase mobile applications or subscription services via a mobile application.

Product Backlog

2.1 Turning requirements into User Stories

ID	Epic	User stories	Acceptance Criteria
1	As a customer, I want to be able to register, create a profile with personal information and receive an email for the registration.	As a customer, I want to create my profile through a signup form.	• User needs to input their name. • User needs to input their email. • User needs to input their contact phone. • User needs to input their bank details. • User needs to input their password. • User needs to input their submit form
2	As a customer, I want to be able to choose from several devices and services that are provided by the company, so that I can choose the one that best suits me.	As a customer, I want to browse and search for packages and deals	• User needs to view all devices and services. • User needs to be able to filter the list based on preference. • User needs to view detailed information of product or service. • User needs to be able to compare the products and services. • User needs to be able to ask for more information for a product.
3	As a customer, I want to be able to make a general or specific enquiry about a specific product or service, so that I can feedback about the service or product.	As a customer, I want to be able to make a general enquiry about the best recommendations via an enquiry form. As a customer, I want to be able to make a specific enquiry about a particular mobile device via an enquiry form.	• User needs to fill in the enquiry form. • User needs to submit the form. • User needs to choose the specific device. • User needs to fill in the enquiry form for the product. • User needs to submit the form.
4	As a customer, I want to be able to see or edit their accounts and make real-time bill payments and be notified of the	As a customer, I want to be able to see my account details. As a customer, I want to be able to edit my account details. As a customer,	• User needs to select his profile. • User needs to edit his profile. • User needs to save the edited details.

	payment confirmation.	I want to be able to make real-time bill payments. As a customer, I want to be notified for the payment confirmation.	• User needs to make real time bill payments. • User needs to be able to choose preferred payment method. • User needs to get notified of the payment confirmation.
5	As a customer, I want to be able to place an order via phoning the sales department, placing it online, or placing it in person, and receive a 15% discount when placing an order through the app.	As a customer, I want to be able to receive a 15% discount when placing an order through the app. As a customer, I want to be able to place an order through the phone call department. As a customer, I want to be able to place an order in person.	• User needs to be able to purchase via the phone department. • User needs to be able to place an order in person. • User needs to be able to place an order via the phone app. • User needs to be able to receive the 15% discount.
6	As a sales representative, I want to be able to profile consumers depending on the programmes they utilise, so that I can sell them products and services tailored to their needs.	As a sales representative, I want to be able to access costumer's profile. As a sales representative, I want to use a user-friendly interface. As a sales representative, I want to be able to view customers' preferred products and services. As a sales representative, I want to be able to view customers' purchase history. As a sales representative, I want to be able to search for customers that have a preferred product or service.	• User needs to be able to access costumer's profile. • User needs to use a user-friendly interface. • User needs to be able to view customers' preferred products and services. • User needs to be able to view customers' purchase history. • User needs to be able to search for customers that have a preferred product or service.

The aforementioned screenshots display the product backlog, which contains EPIC tales that we have turned into user stories and their associated acceptance criteria. These graphic tools shed light on the design procedure and demonstrate how we successfully converted high-level objectives into doable tasks.

2.2 User stories prioritisation with MOSCOW (Epics Breakdown)

ID	User story	MOSCOW
1.	As a customer, I want to create my profile through a signup form.	Must have
2.	As a customer, I want to browse and search for packages and deals	Should have
3.	As a customer, I want to be able to make a general enquiry about the best recommendations via an enquiry form.	Could have
4.	As a customer, I want to be able to see my account details.	Must have
5.	As a customer, I want to be able to receive a 15% discount when placing an order through the app.	Could have
6.	As a sales representative, I want to be able to search for customers that have a preferred product or service.	Could have
7.	As a sales representative, I want to use a user-friendly interface.	Must have
8.	As a sales representative, I want to be able to view customers' preferred products and services.	Must have
9.	As a sales representative, I want to be able to view customers' purchase history.	Must have
10.	As a customer, I want to be able to make a specific enquiry about a particular mobile device via an enquiry form.	Should have
11.	As a sales representative, I want to be able to access costumer's profile.	Must have
12.	As a customer, I want to be able to edit my account details.	Must have
13.	As a customer, I want to be able to place an order through the phone call department.	Could have
14.	As a customer, I want to be able to place an order in person.	Should have
15.	As a customer, I want to be notified for the payment confirmation.	Should have
16.	As a customer, I want to be able to make real-time bill payments.	Must have

2.3 Minimum Viable Product

After careful consideration on the prioritization using the Moscow Prioritization, the Minimum Viable Product (MVP) is the "Must Have's" as they are important for the initial release of the product. The "Should Have's" and "Could Have's" can be implemented if time allows. The "Won't Have's" should be degraded because they consume a lot of resources and should be left for later updates.

2.4 User stories estimation using Planning Poker

		Votes				
ID	User stories	Marini	Dimitrios	Oviyan	Tejesh	Anayatul
1	As a customer, I want to create my profile through a signup form.	3	5	3	3	5
2	As a customer, I want to browse and search for packages and deals	5	5	5	5	5
3	As a customer, I want to be able to make a general enquiry about the best recommendations via an enquiry form.	3	3	3	3	3
4	As a customer, I want to be able to see my account details.	3	3	3	3	3
5	As a customer, I want to be able to receive a 15% discount when placing an order through the app.	5	5	5	5	5
6	As a sales representative, I want to be able to search for customers that have a preferred product or service.	8	8	8	8	8
7	As a sales representative, I want to use a user-friendly interface.	5	5	5	3	3
8	As a sales representative, I want to be able to view customers' preferred products and services.	5	8	5	5	5
9	As a sales representative, I want to be able to view customers' purchase history.	5	5	8	5	5
10	As a customer, I want to be able to make a specific enquiry about a particular mobile device via an enquiry form.	3	3	3	3	3
11	As a sales representative, I want to be able to access costumer's profile.	8	8	8	8	8
12	As a customer, I want to be able to edit my account details.	3	3	3	5	3
13	As a customer, I want to be able to place an order through the phone call department.	5	5	5	5	5
14	As a customer, I want to be able to place an order in person.	5	5	5	5	5
15	As a customer, I want to be notified for the payment confirmation.	3	5	3	3	3
16	As a customer, I want to be able to make real-time bill payments.	5	5	5	5	5

Sprint One

3.1 Sprint Planning and Sprint Backlog

3.1.1 Sprint Plan

Sprint 1 (5 Days):
Allow a customer to register and create a profile with relevant information.
Customers can browse and search for packages or deals.
Customers can make a general enquiry or an enquiry about a specific package or deal.
Implement email confirmation for new registrations and orders.
Develop the user interface and design for the mobile app.

3.1.2 Sprint Goal

To provide clients with a user-friendly mobile app that allows them to register, establish a profile, browse, and search for packages and discounts, make general or package-specific inquiries, and receive email confirmations for new registrations and orders. By enhancing the app's user interface and design, we also hope to increase its appeal and usability.

3.1.3 Sprint “Definition of Done”

1. Both the sign-up form and the option for creating a profile have been thoroughly tested.
2. The search option is user-friendly, and users can easily browse and search for packages or deals.
3. Users have the option to claim deals, and the software appropriately applies automatic savings.
4. Users receive confirmation emails automatically for new memberships and orders.
5. The user interface and design have been enhanced to make them more appealing and intuitive.
6. The coding standards are followed, and every code is adequately documented.
7. The acceptance criteria established throughout the sprint planning process have been met by all features and functionalities after testing.

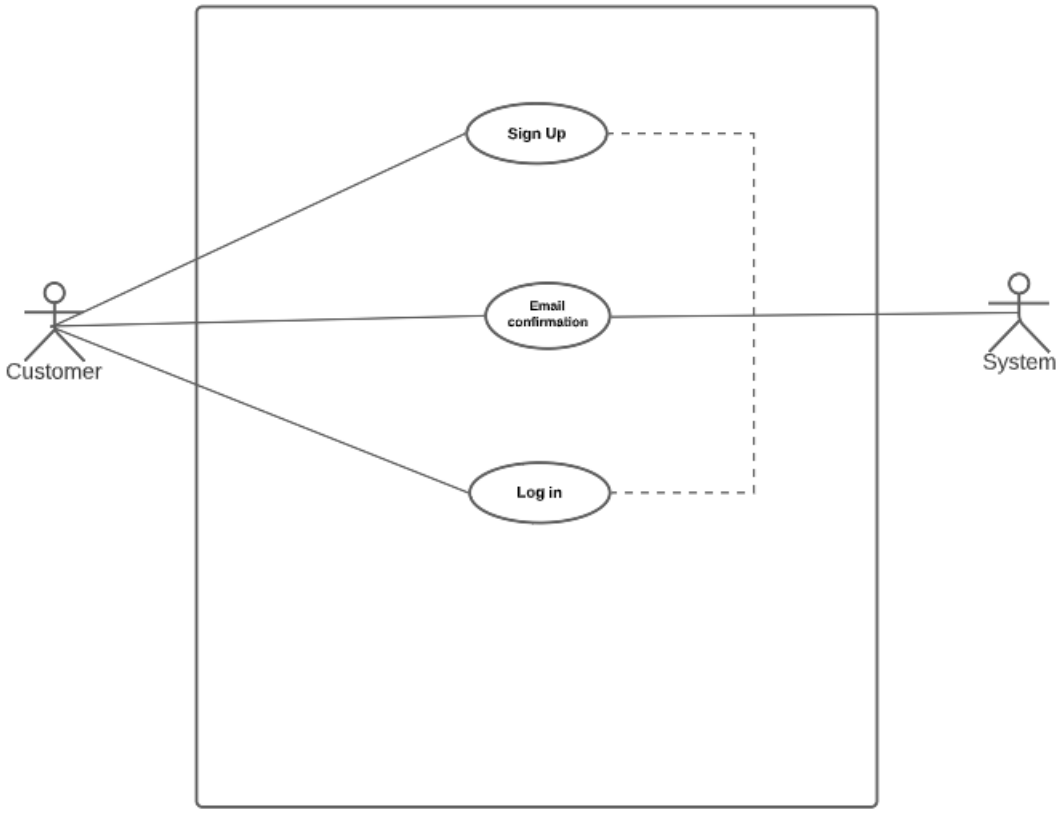
3.1.4 Sprint Backlog

User Story	Task	Day estimation
Allow a customer to register and create a profile with relevant information.	Build a registration form user interface with fields for personal data like name, email, password, and contact information. Provide the capability for a database to hold user data. For new registrants, implement email confirmation.	3
Customers can browse and search for packages or deals.	Provide a user interface that allows users to browse and search for bundles and specials. Provide a search box so that clients may look for particular bundles or bargains. Provide a filtering system that would enable clients to arrange offers and bundles according to factors like price, date, and location. Provide a method for showing pertinent data for each bundle or offer.	5
Customers can make a general enquiry or an enquiry about a specific package or deal.	Provide a user interface that allows users to enquire about general inquiries or specific packages or deals. Create a form that consumers may use to contact you. Create a backend system to take inquiries and store them in a database. Provide a mechanism for notifying personnel when a new query is received.	4
Implement email confirmation for new registrations and orders.	Create a system that will send clients' confirmation emails after they register or place a purchase. Add pertinent facts in the confirmation email, such as account information, an overview of the order, and payment authorization.	3
Develop the user interface and design for the mobile app.	For the user interface of the mobile app, create wireframes. Based on the wireframes, create the user interface for the mobile app. Make sure the mobile app design is aesthetically beautiful, accessible, and user-friendly.	5

In the screenshot above, we have showcased the sprint backlog for the first sprint, where we have shown the tasks for each user story and a day estimate that we used to project the length of the sprint.

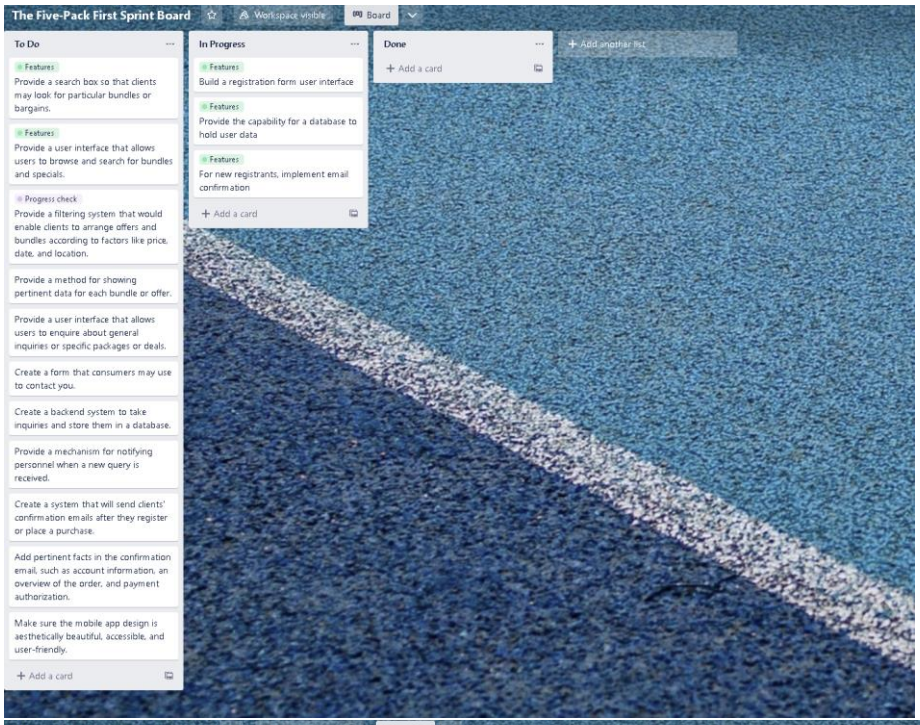
3.2 Sprint Cycle

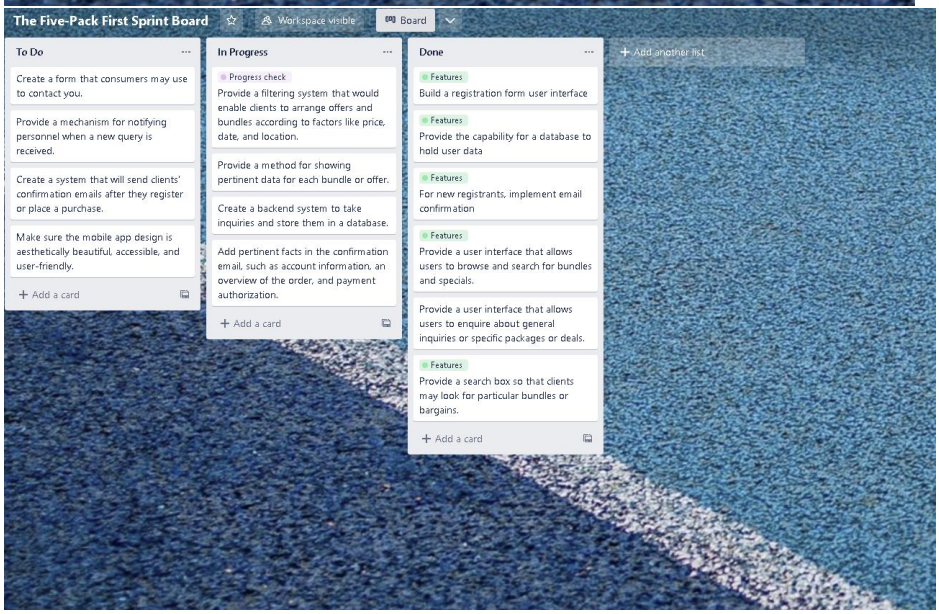
3.2.1 Use-Case Diagram

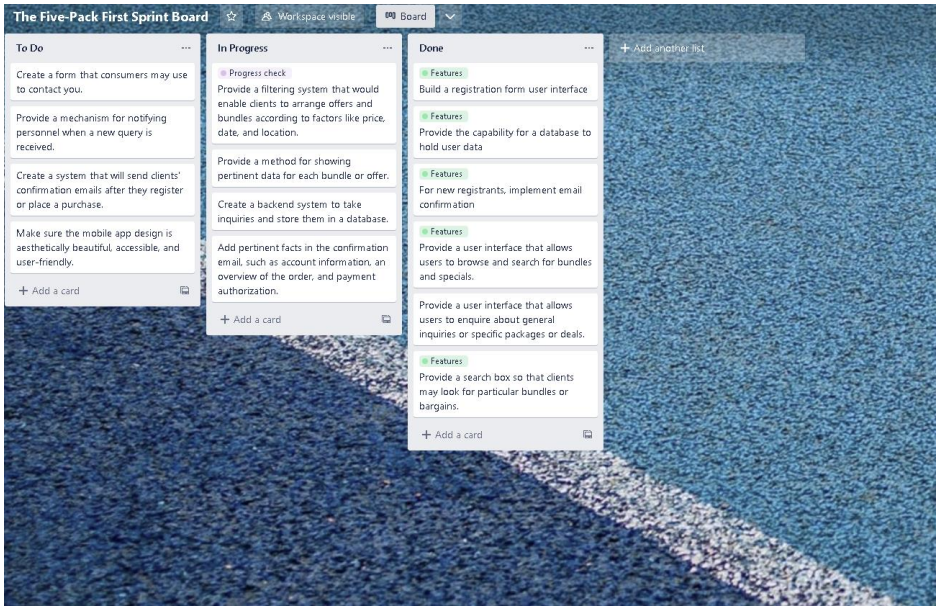


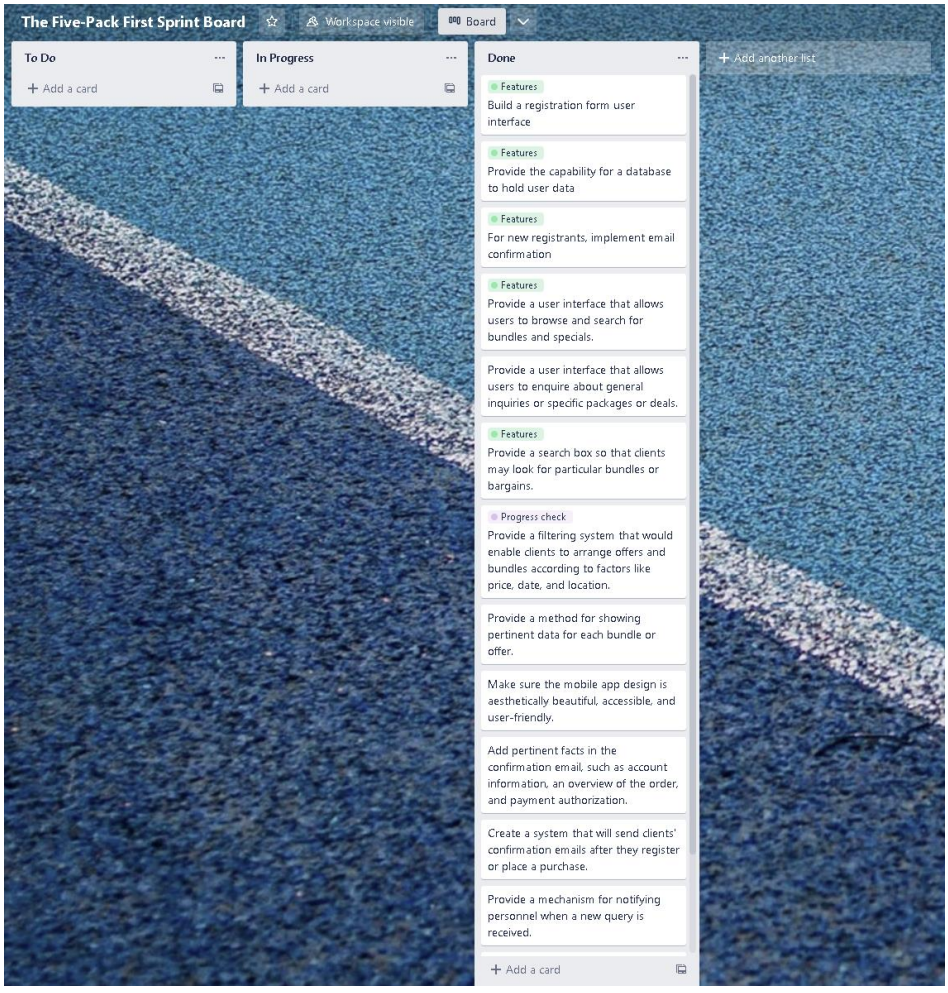
In the screenshot above, we have showcased the use case diagram that represents our system and what we have implemented in the first sprint, as well as its stakeholders and the tasks that the system offers.

3.2.2 Scrum Board





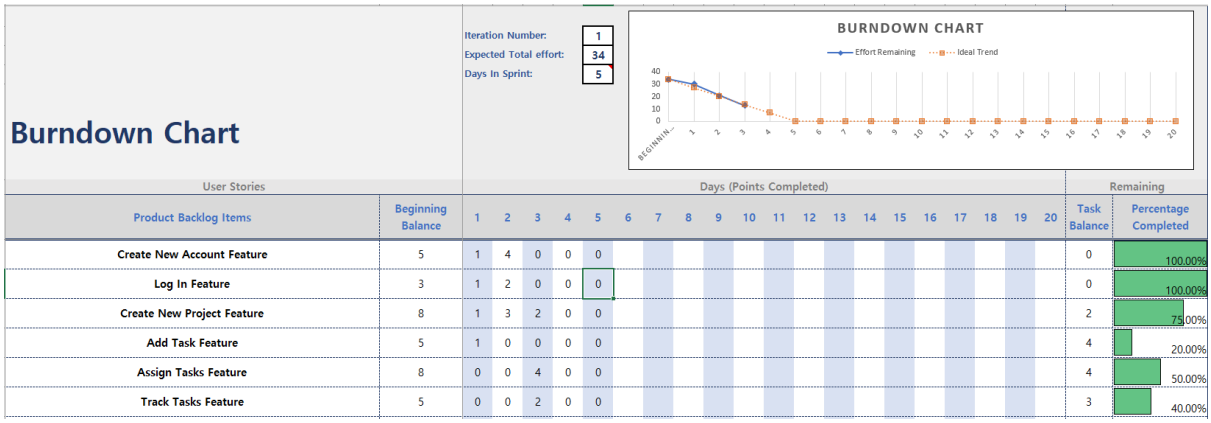




In the screenshots above, we have showcased the scrum board that we utilised in our first sprint to be able to distinguish the tasks and be able to position them based on whether they were to be done, in progress or done.

3.2.3 Burndown Chart

3.2.3.1 Halfway



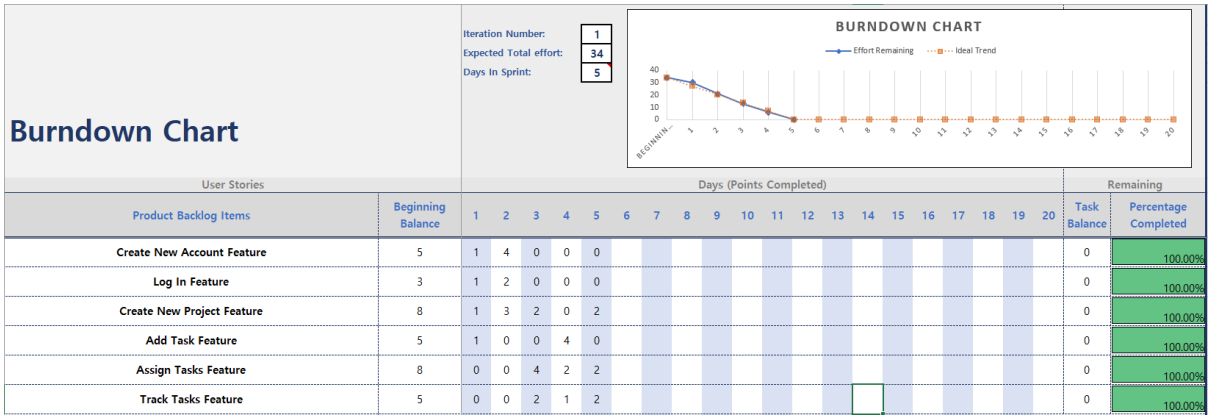
Burndown Chart

Iteration Number: 1
Expected Total effort: 34
Days In Sprint: 5

BURNDOWN CHART
Effort Remaining
Ideal Trend

User Stories		Days (Points Completed)																				Remaining	
Product Backlog Items	Beginning Balance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Task Balance	Percentage Completed
Create New Account Feature	5	1	4	0	0	0																0	100.00%
Log In Feature	3	1	2	0	0	0																0	100.00%
Create New Project Feature	8	1	3	2	0	0																2	75.00%
Add Task Feature	5	1	0	0	0	0																4	20.00%
Assign Tasks Feature	8	0	0	4	0	0																4	50.00%
Track Tasks Feature	5	0	0	2	0	0																3	40.00%

3.2.3.2 Full



Burndown Chart

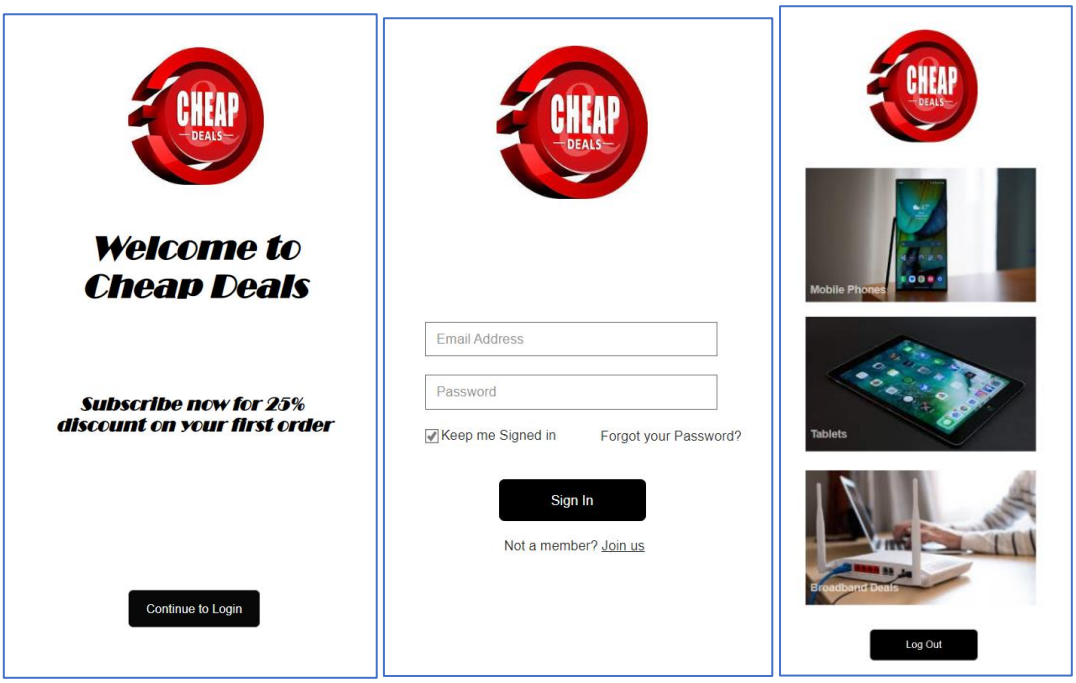
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BURNDOWN CHART
Effort Remaining
Ideal Trend

User Stories		Days (Points Completed)																				Remaining	
Product Backlog Items	Beginning Balance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Task Balance	Percentage Completed
Create New Account Feature	5	1	4	0	0	0																0	100.00%
Log In Feature	3	1	2	0	0	0																0	100.00%
Create New Project Feature	8	1	3	2	0	2																0	100.00%
Add Task Feature	5	1	0	0	4	0																0	100.00%
Assign Tasks Feature	8	0	0	4	2	2																0	100.00%
Track Tasks Feature	5	0	0	2	1	2																0	100.00%

In the screenshots above, we have showcased the burndown charts that we utilised in our first sprint to be able to see the completion of our sprint in contrast to our projection.

3.2.4 Prototype



In the screenshots above, we have showcased three pages of the prototype (subscription, logging in and main menu).

3.2.5 Sprint Daily Meeting

Duration of Sprint: 5 Days

Sprint Start: 6th March

Sprint End: 10th March

3.2.6 Sprint Review

The Five Pack's Team:

Agenda:

Creating a new account and logging into an existing one through user registration

Dashboard page displaying user tasks and projects.

Function to establish a new project and add tasks to it.

the capacity to delegate work to teammates and monitor their progress.

Stakeholders provide comments and pose questions:

A functionality to establish project deadlines and show them on the dashboard was proposed by the project manager.

The client asked that tasks on the dashboard be able to be filtered by priority or due date.

The Team and Product Owner evaluate and rank the submitted comments.

The team will look into the viability of adding task filtering capability and is dedicated to introducing functionality to establish project deadlines. Product owners provide input and priorities to the product backlog.

The next Sprint's purpose is to significantly enhance platform user experience by introducing new features and addressing identified defects, which is agreed upon by the team and stakeholders.

3.2.7 Sprint Retrospective

Date: 3 March 2023

Tejesh, Oviyan, Anayat, Dimitris, and Marini

Agenda:

What aspects of the Sprint went well?

What might have gone more smoothly?

What will we change for the following time?

Any further suggestions or remarks?

Discussions:

What aspects of the Sprint went well? The lyrics are intended to convey that we succeeded in finishing all of the features we had for this sprint. The staff works well together and has outstanding communication, which enables us to handle issues quickly and effectively.

Tejesh: In order to keep everyone informed of our progress and any challenges we were encountering; the daily scrum sessions were tremendously beneficial. Also, by seeing issues early on, we can work to resolve them.

Anayat: The group is really helpful and always willing to lend a hand. We were able to continue working efficiently and finish all the features on schedule.

Dimitrios: Because of how well-organized the team was, we were able to complete each of our unique jobs without difficulty or misunderstanding. We finished the sprint on schedule as well.

Marini: It is remarkable how dedicated and committed the crew is to produce a high-quality product. In order to accomplish our objectives, we were able to collaborate and work productively. What might have gone more smoothly?

Oviyan: The dashboard describes how we occasionally had to wait for someone to finish their task before moving on. To prevent any delays, we should aim to collaborate more closely the next time. When the Sprint ended, we could have conducted further testing to make sure everything was functioning properly.

Anayat: To make sure we were all on the same page, we need to have spent more time talking about design and user experience.

Tejesh: To prevent ambiguity or overlap, we might clarify the Sprint's objectives and each of our specific responsibilities.

Marini: We might have given unknown problems that could come up during the race more time.

What will we change for the following time? To prevent any delays, we will collaborate more closely and communicate more frequently, according to the lyrics.

Oviyan: To ensure that everything functions as anticipated, we will concentrate on additional testing.

Anayat: To make sure we're all on the same page, we'll spend more time discussing user experience and design.

Dimitrios: To prevent confusion or job overlap, we will more clearly outline the Sprint's scope and each of our particular responsibilities.

Marini: We'll spend more effort dealing with unanticipated problems that could come up during the sprint.

Any further suggestions or remarks?

Tejesh: Ultimately, I think we did a terrific job, and I'm happy of what we accomplished in this race, according to the lyrics.

Oviyan: I concur and am forward to collaborate with this team once more in the future.

Anayat: I'm grateful for everyone's efforts and attention, and I've gained a lot from this encounter.

Dimitrios: I am really appreciative to have had the chance to work with such a bright and committed group of people.

Marini: The team's dedication and the calibre of the work we produced amazed me. We will continue to succeed in new ventures, I have no doubt.

Sprint Two

4.1 Sprint Planning and Sprint Backlog

4.1.1 Sprint Plan

Sprint 2 (4 Days):
Allow customers to customize their own package based on their mobile and/or broadband usage.
Implement the Double Package and Triple Package deals.
Customers can view and amend their accounts.
Customers can pay their bills in real-time.
Develop and integrate the new CRM system.

4.1.2 Sprint Goal

At the end of this sprint, customers will be able to fully personalize their own packages, adjust their accounts, make real-time payments, and fully utilize and execute the new Double Package and Triple Package offers. Furthermore, a new CRM system will be built to increase client service and retention.

4.1.3 Sprint “Definition of Done”

1. Customers are able to easily customize their own package based on their own mobile devices and broadband usage.
2. Double Package and Triple Package deals are correctly implemented and tested, whereby allowing customers to take full advantage of these deals.
3. Customers are able to view and make changes to their account information, such as personal details, usage history or billing information.
4. Allows customers to pay bill in real time, meaning the development have to integrate a fast payment process onto the platform, which is without bugs or delays.

5. The new fully developed CRM system has to have all the relevant data to relay information to customers.
6. The new system has to be thoroughly tested, where all the problems have to be solved before the initial release.
7. There has to be a relevant documentation created, including user guidelines.

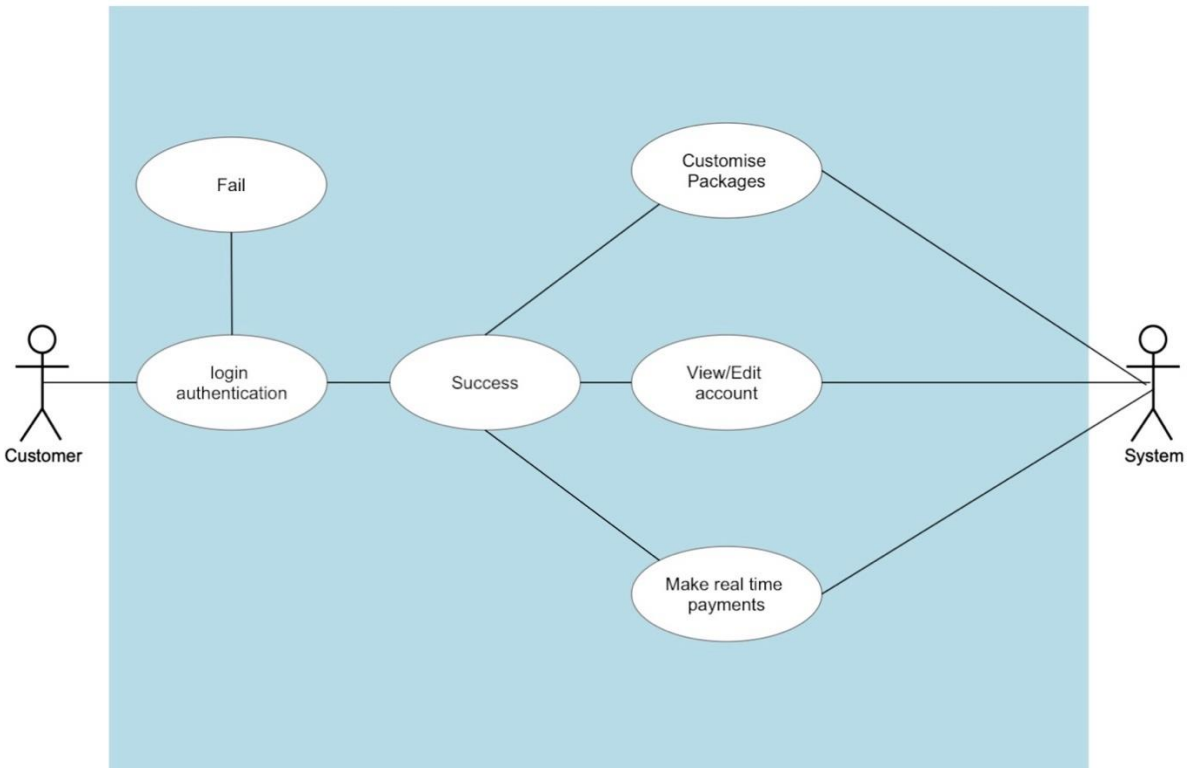
4.1.4 Sprint Backlog

Sprint Backlog	Task	Day estimation
Implement login and authentication feature	Implementing the login form and the user interface components which are required for authentication. Designing the user authentication, which includes a password policy, or a password reset. Integrating the login and authentication with the systems frontend and backend. Develop test to make sure that the system is working.	2
Develop user profile page	Designing the user interface for the profile page and implementing frontend and backend components. Ability to make changes to the profile. Integrating the frontend and backend together and testing out whether the system is ready or not.	3
Create product catalogue page	Designing the user interface for the product catalogue page and implementing frontend and backend components. Developing test to test if the system is responsive or not.	3
Build shopping cart functionality	Designing the user interface for the shopping cart page and implementing frontend and backend components. Creating calculation function. Ability to save item in the cart and recalling them whenever required. Ability to make payment directly from the cart. Developing test to test if the system is responsive or not.	3

In the screenshot above, we have showcased the sprint backlog for the second sprint, where we have shown the tasks for each user story and a day estimate that we used to project the length of the sprint.

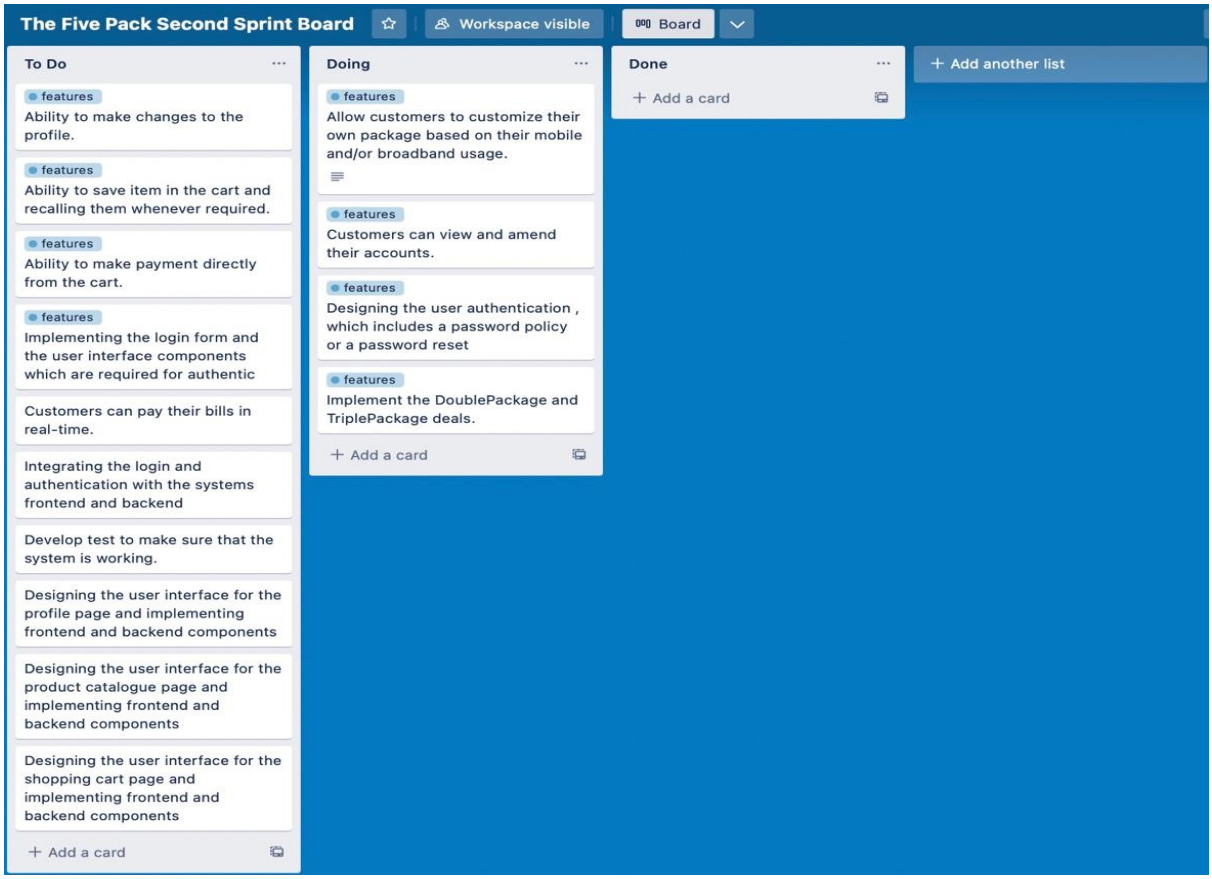
4.2 Sprint Cycle

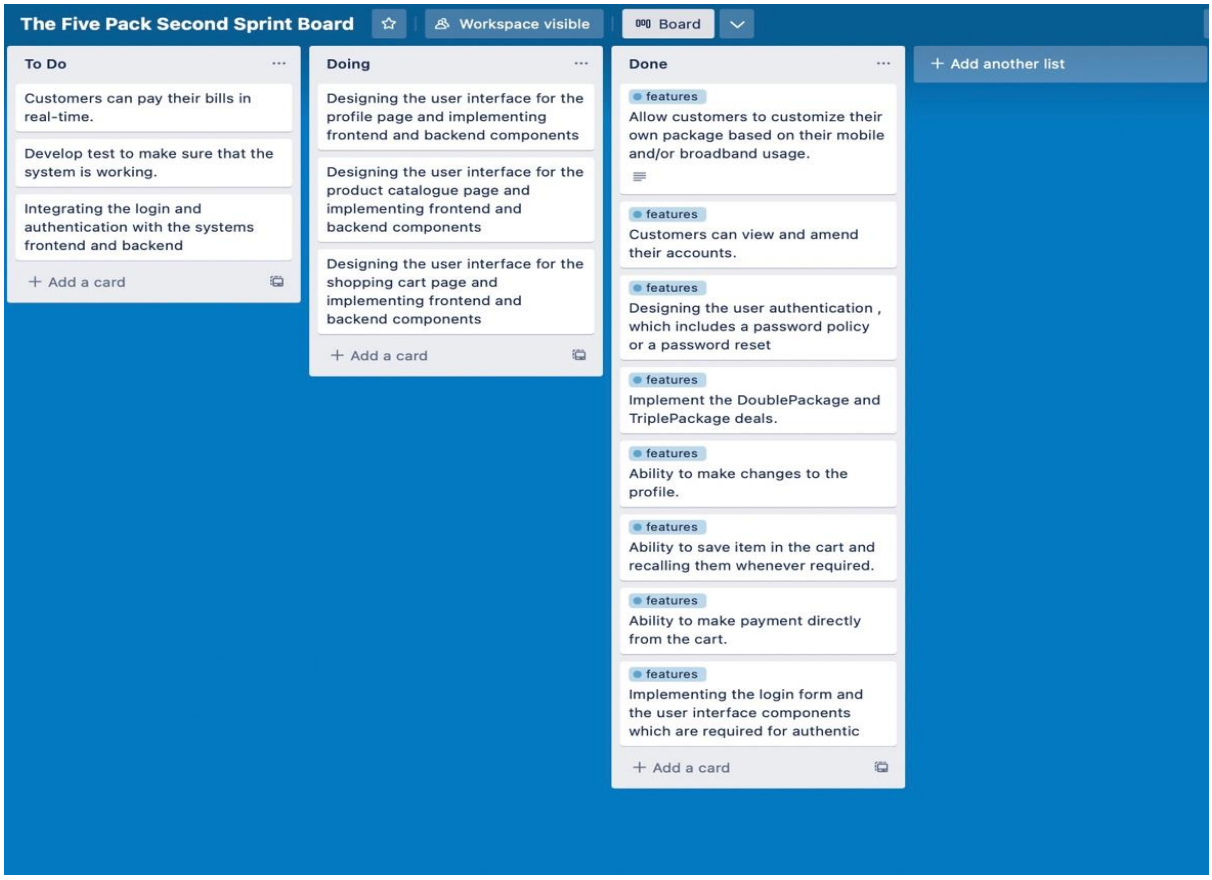
4.2.1 Use-Case Diagram

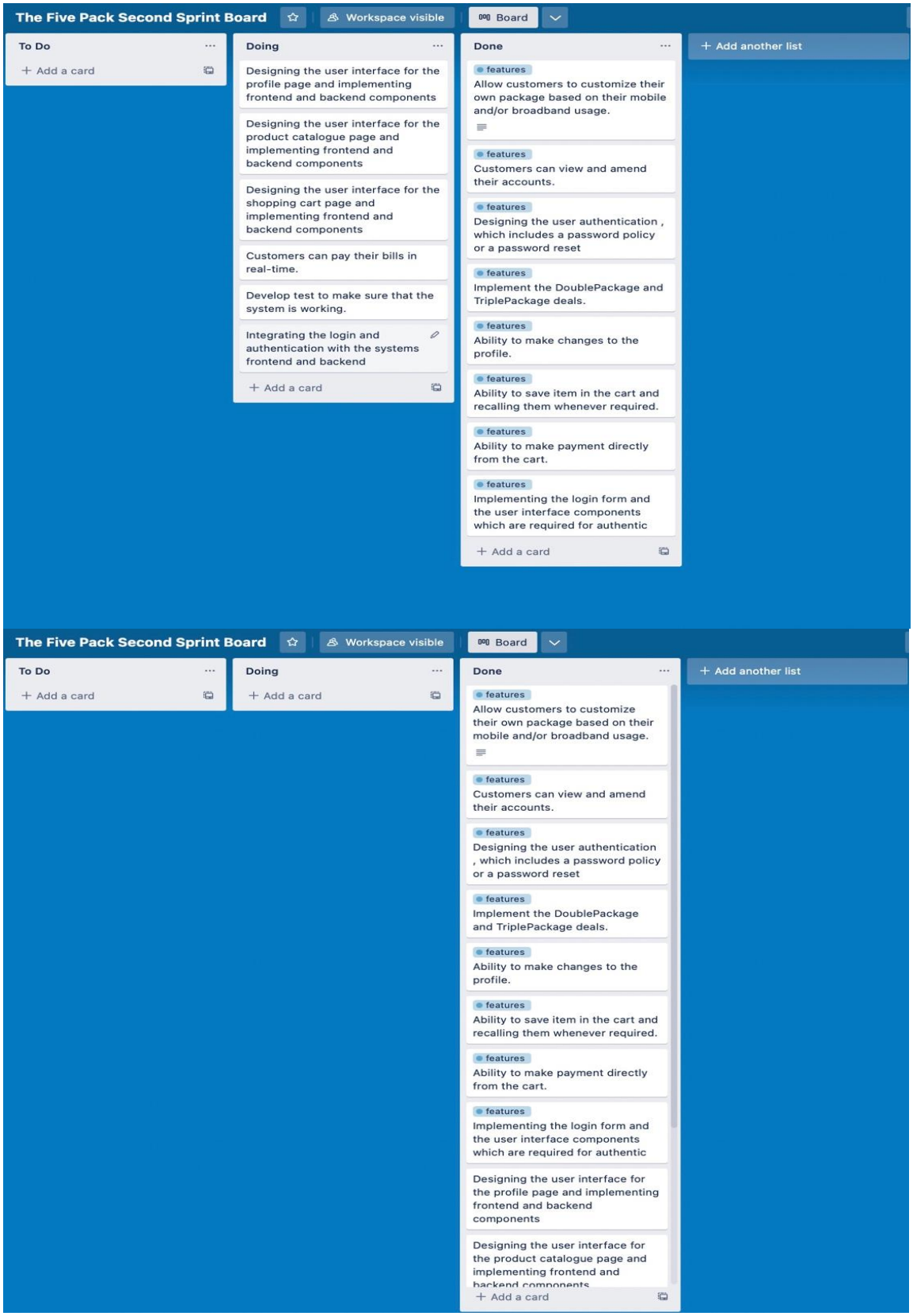


In the screenshot above, we have showcased the use case diagram that represents our system and what we have implemented in the second sprint, as well as its stakeholders and the tasks that the system offers to the customer.

4.2.2 Scrum Board



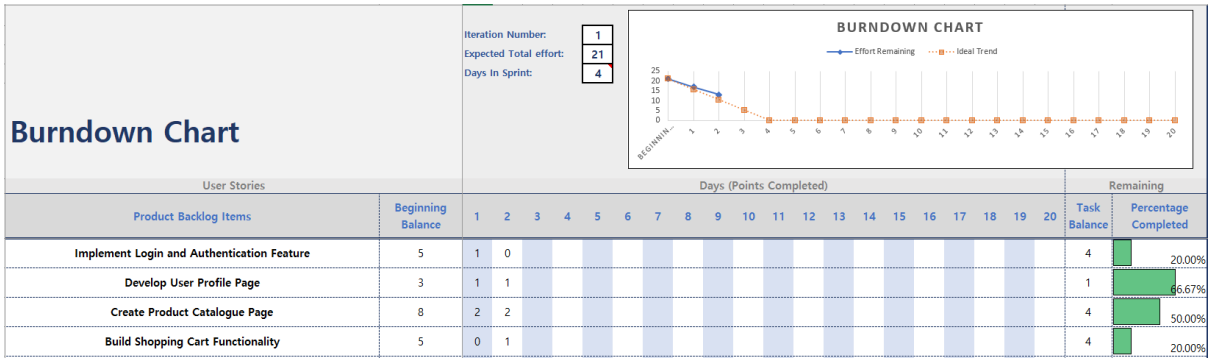




In the screenshots above, we have showcased the scrum board that we utilised in our second sprint to be able to distinguish the tasks and be able to position them based on whether they were to be done, in progress or done.

4.2.3 Burndown Chart

4.2.3.1 Halfway

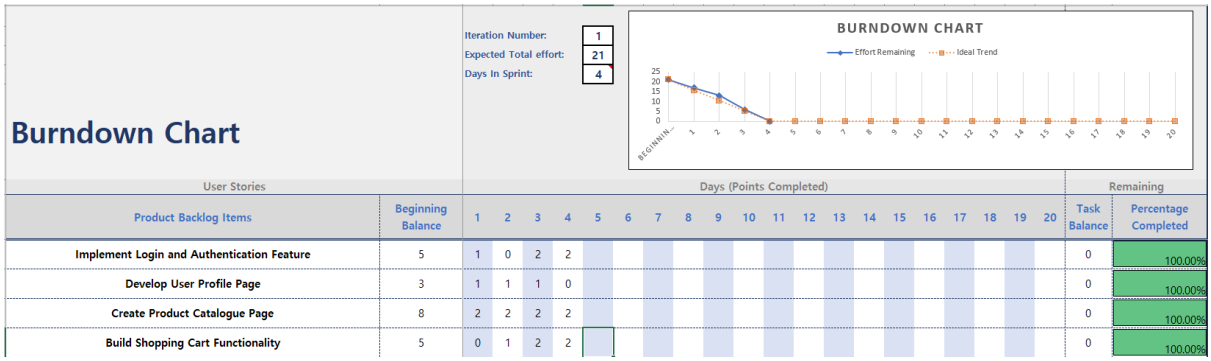


Burndown Chart

Iteration Number: 1
Expected Total effort: 21
Days In Sprint: 4

User Stories		Days (Points Completed)																				Remaining	
Product Backlog Items	Beginning Balance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Task Balance	Percentage Completed
Implement Login and Authentication Feature	5	1	0																			4	20.00%
Develop User Profile Page	3	1	1																			1	66.67%
Create Product Catalogue Page	8	2	2																			4	50.00%
Build Shopping Cart Functionality	5	0	1																			4	20.00%

4.2.3.2 Full



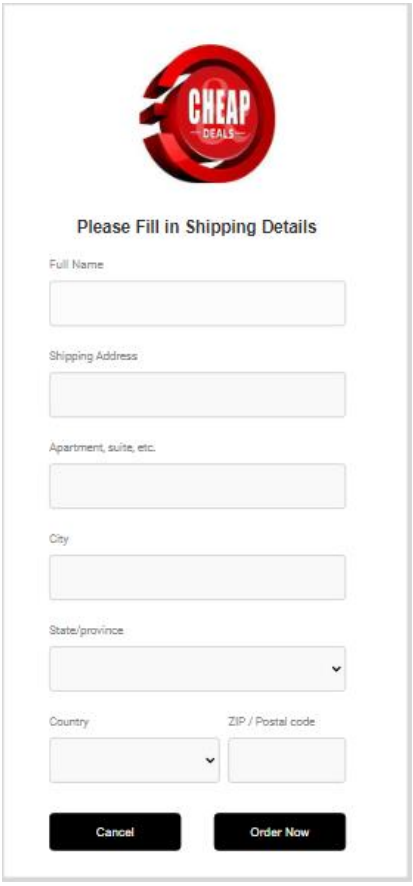
Burndown Chart

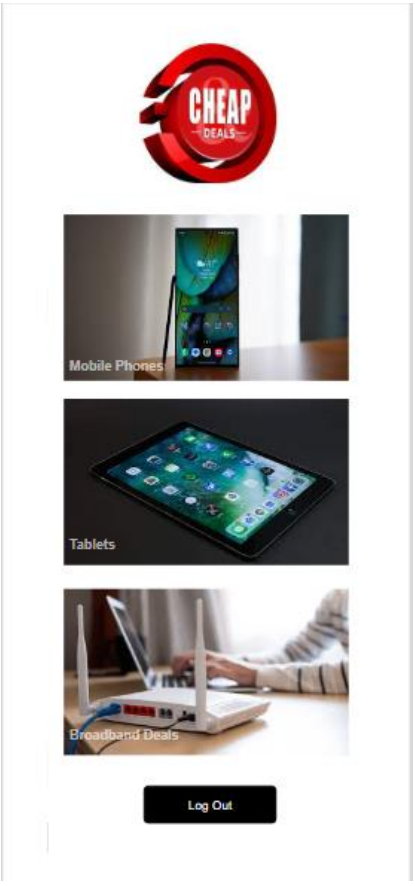
Iteration Number: 1
Expected Total effort: 21
Days In Sprint: 4

User Stories		Days (Points Completed)																				Remaining	
Product Backlog Items	Beginning Balance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Task Balance	Percentage Completed
Implement Login and Authentication Feature	5	1	0	2	2																	0	100.00%
Develop User Profile Page	3	1	1	1	0																	0	100.00%
Create Product Catalogue Page	8	2	2	2	2																	0	100.00%
Build Shopping Cart Functionality	5	0	1	2	2																	0	100.00%

In the screenshots above, we have showcased the burndown charts that we utilised in our second sprint to be able to see the completion of our sprint in contrast to our projection.

4.2.4 Prototype





In the screenshots above, we have showcased the shipping details form and the optional menu of subscribed users.

4.2.5 Sprint Daily Meeting

Duration of Sprint: 4 Days

Sprint Start: 13th March

Sprint End: 16th March

4.2.6 Sprint Review

With the help of each team member's contribution to the allocated work, the team has overall achieved good progress throughout this sprint. The group worked on a number of features, such as user authentication and login, user profile pages, product category pages, task filtering, pagination, sorting, and reporting functions.

The team did face a few challenges during this sprint, though. On a user's profile page, Tejesh had successfully finished his filtering task without any difficulties, Oviyan had trouble developing the form entries and had to specify some requirements for the pagination capability. Clarifying the task filtering criteria will let Anayat link its backend reporting capabilities. Dimitrios had several delays with the search function, and it was challenging to create the sort of function. While she was awaiting API documentation from the backend team, Marini's work was delayed.

Despite these obstacles, the team was able to address these issues and come up with solutions by working together. Customers also provide input, and they assess how the project is coming along.

Overall, the team worked well together and communicated effectively during this sprint, which allowed them to advance towards their objectives. To make sure that subsequent sprints are successful, the team must keep working to remove bottlenecks and improve communication.

4.2.7 Sprint Retrospective

A number of features and functionality advanced during the sprint. The group collaborated well, and everyone continued to work towards the same goal. Throughout the sprint phase, there are a few obstacles to go past, though.

The team shared their accomplishments and day's goals during the initial meeting. There weren't any big problems back then. Although Anayat wanted to explain the design requirements for the product category page, Oviyan battled with the layout of the form entries on the user profile page. Dimitrios is anticipating the front-end team's answer to his API queries for cart functionality. Marini recommended that we talk about these issues after the meeting.

Everyone provided the team with an update on their progress and the challenges they faced at the second meeting. Oviyan struggled with the pagination specifications, but Tejesh successfully finished the job filtering. Dimitrios' search engine is running a little slowly, and Anayat is having difficulties integrating the reporting features on the backend. Marini informed the group that they were behind schedule because the backend team needed to provide API documentation.

The team provided an update on their work and any problems they ran across at the third meeting. After obtaining the API documentation from the backend team, Tejesh finished the job filtering. Task dashboard functionality was finished by Oviyan, Anayat addressed a bug with task progress monitoring and reporting, and Dimitrios enhanced the search feature's functionality. By using progress reports, Marini updated the customer after evaluating the development.

The group addressed its progress and day's plans at the fourth meeting. Tejesh finished job pagination and is now concentrating on task filtering. The task export feature has been completed, and Oviyan is presently working on the task sorting function. As Dimitrios worked on the task filtering and task sorting feature, Anayat finished the task dashboard functionality and worked on the task user profile page. Marini assessed the findings and provided the group with feedback.

The team provided an update on their work and any problems they ran across during the most recent meeting. Tejesh concentrated on task filtering, Oviyan worked on task sorting, Anayat had trouble with the layout of the user profile page, and Dimitrios had trouble with sorting. After the meeting, Marini advised resolving such issues.

Overall, the team accomplished a lot throughout the Sprint. However, some problems, such those with design and implementation, need to be solved. The team may endeavour to enhance communication in subsequent sprints and make sure that everyone is aware of the specifications and demands for each feature.

Sprint Three

5.1 Sprint Planning and Sprint Backlog

5.1.1 Sprint Plan

Sprint 3 (5 Days):
Offer special deals directly through the mobile app by profiling customers.
Improve customer service by reducing waiting time for CSR.
Integrate real-time assistance on the online portal.
Improve customer satisfaction by improving the information available to the CSRs.
Test and debug the mobile app and CRM system.
Finalize user documentation and training materials.

5.1.2 Sprint Goal

Our final sprint goal is to leave a lasting positive impression on our customers by delivering enhancements to the mobile app. Our focus is on providing a seamless customer experience and maximizing their satisfaction. We aim to finish strong by ensuring all outstanding issues are resolved and our product is ready to exceed customer expectations.

5.1.3 Sprint "Definition of Done"

1. The feature for offering special deals directly through the mobile app by profiling customers has been implemented, tested, and accepted by the product owner.
2. Customer service processes have been optimized to reduce waiting time for CSR, and the changes have been tested and approved by the product owner.
3. Real-time assistance has been integrated into the online portal, tested, and accepted by the product owner.
4. The quality of the information available to the CSRs has been improved, tested, and approved by the product owner.
5. The mobile app and CRM system have been thoroughly tested, and all identified bugs and issues have been resolved.
6. User documentation and training materials have been updated, reviewed, and approved by the product owner.

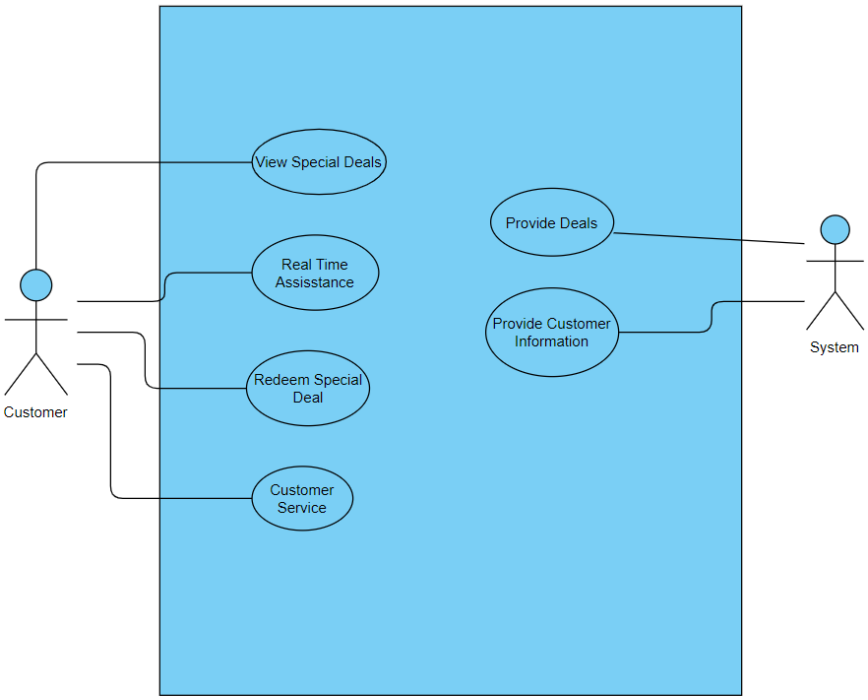
5.1.4 Sprint Backlog

Sprint Backlog	Task	Day estimation
Offer special deals directly through the mobile app by profiling customers.	Designing the algorithm for profiling customers that can get the available offers and implementing frontend and backend components. Developing test to test if the system is responsive or not.	3
Improve customer service by reducing waiting time for CSR.	Designing the user interface CSR and implementing frontend and backend components. Developing test to test if the system is responsive or not.	1
Integrate real-time assistance on the online portal.	Designing and integrating an assistance method for the customer to use in real time.	4
Improve customer satisfaction by improving the information available to the CSRs.	Finding areas where customer satisfaction may drop and developing and implementing plans to solve the problematic areas.	1
Test and debug the mobile app and CRM system.	Create a testing plan, followed by finding error and formulating a plan to debug the errors to correct the system.	5
Finalize user documentation and training materials.	Document the entire process to finalize the reports. Create a training manual for the employees to follow and understand.	3

In the screenshot above, we have showcased the sprint backlog for the third and final sprint, where we have shown the tasks for each user story and a day estimate that we used to project the length of the sprint.

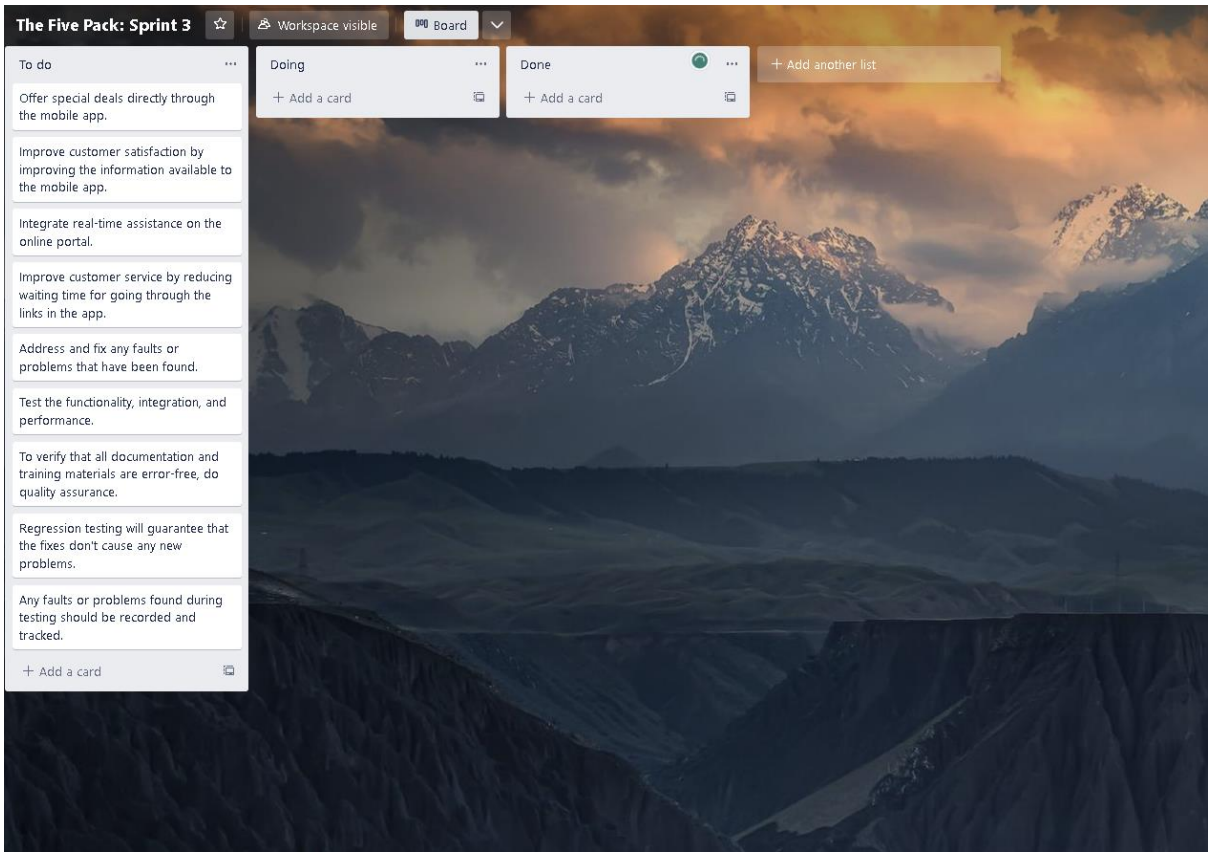
5.2 Sprint Cycle

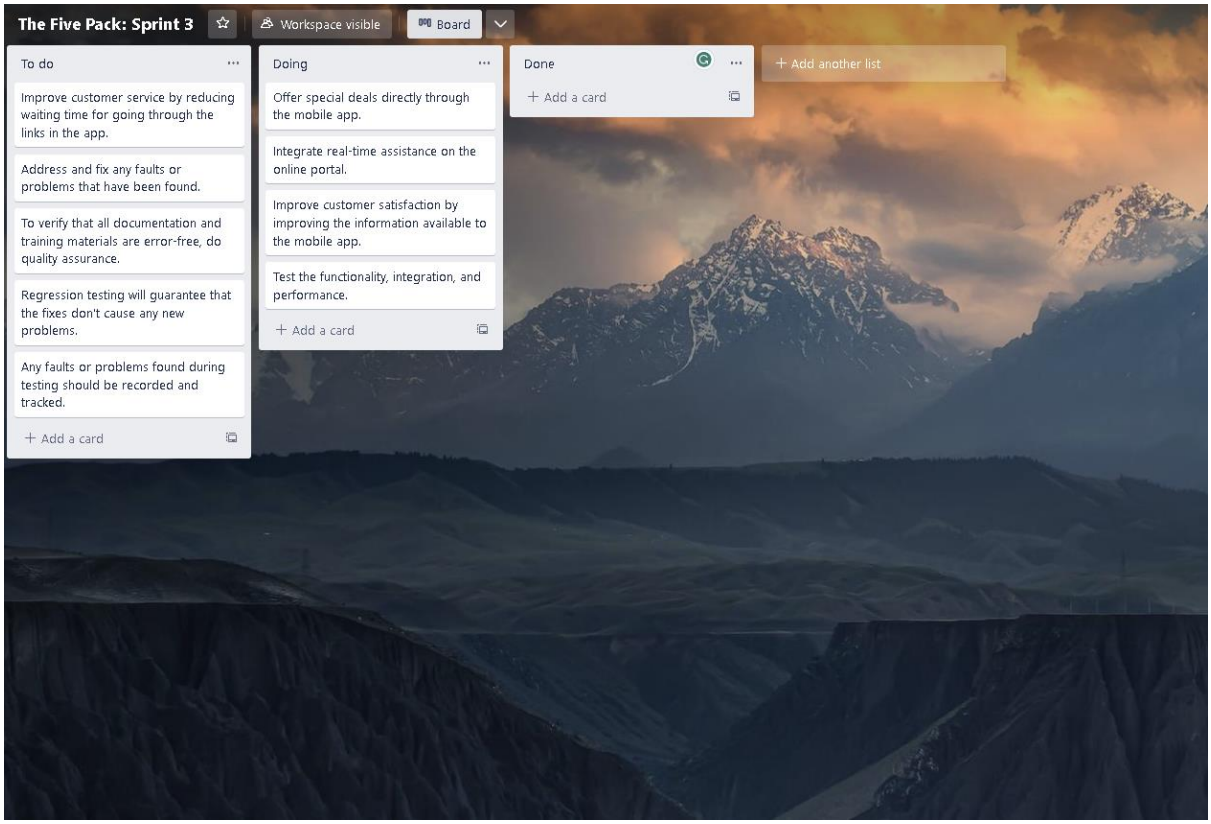
5.2.1 Use-Case Diagram

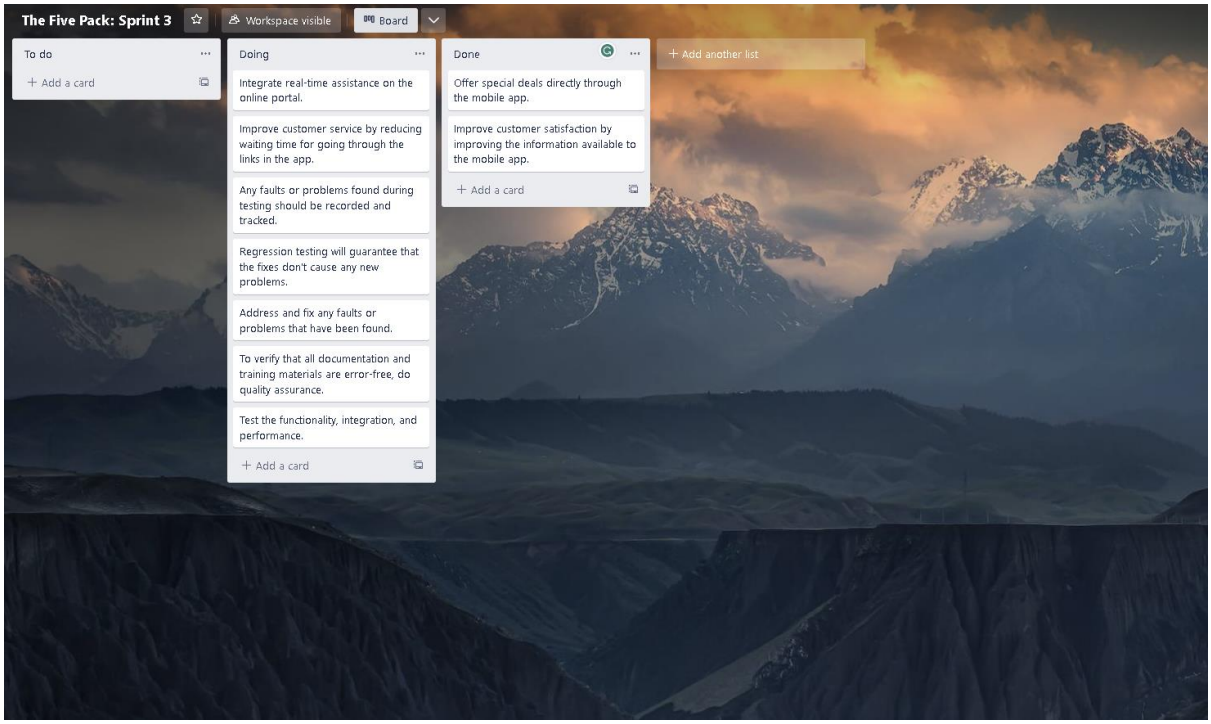


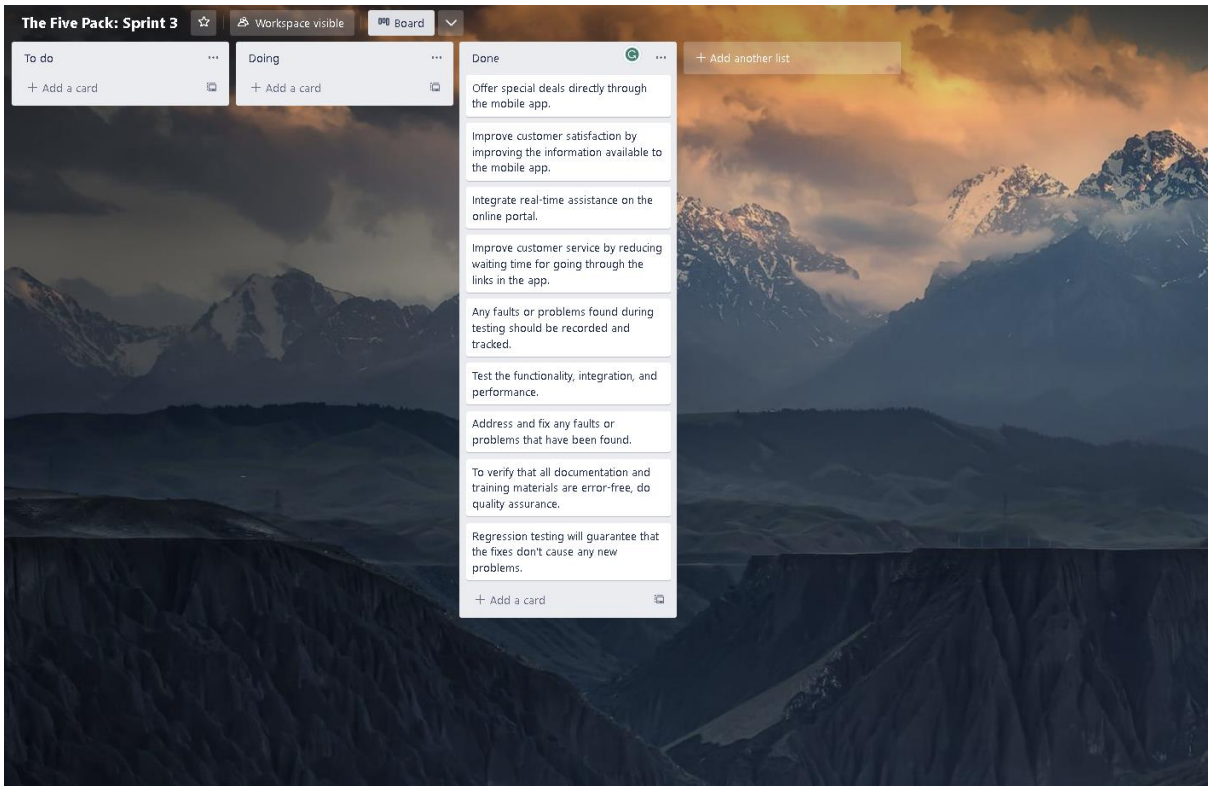
In the screenshot above, we have showcased the use case diagram that represents our system and what we have implemented in the third sprint, as well as its stakeholders and the tasks that the system offers to the customer.

5.2.2 Scrum Board





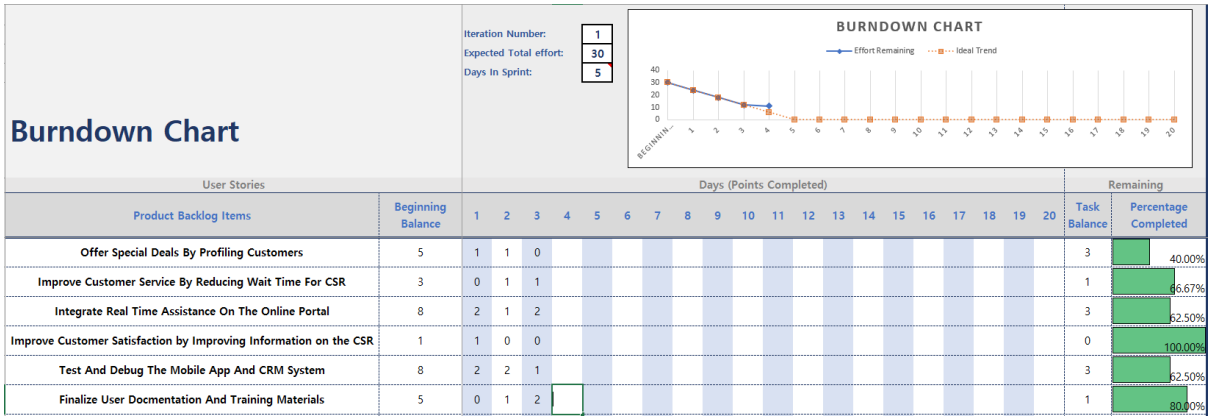




In the screenshots above, we have showcased the scrum board that we utilised in our second sprint to be able to distinguish the tasks and be able to position them based on whether they were to be done, in progress or done.

5.2.3 Burndown Chart

5.2.3.1 Halfway

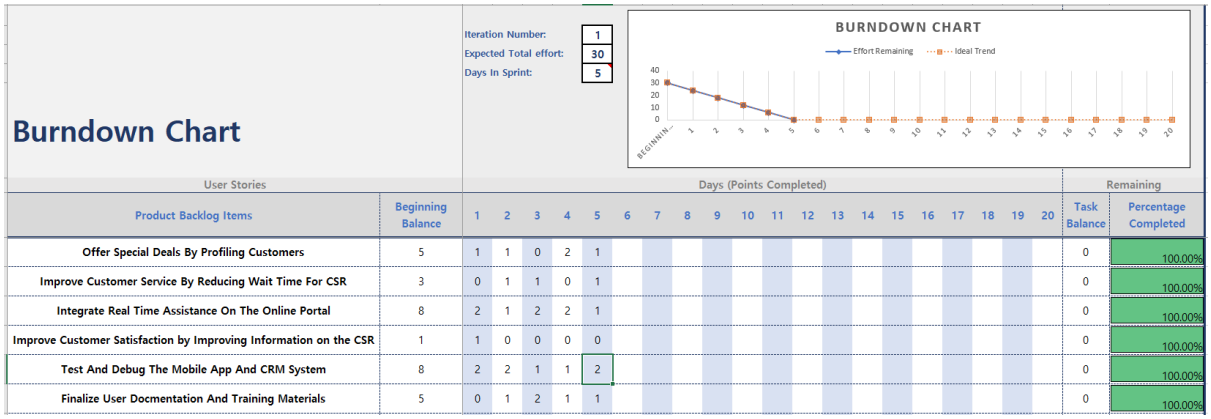


Burndown Chart

Iteration Number: 1
Expected Total effort: 30
Days In Sprint: 5

User Stories		Days (Points Completed)																				Remaining	
Product Backlog Items	Beginning Balance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Task Balance	Percentage Completed
Offer Special Deals By Profiling Customers	5	1	1	0																		3	40.00%
Improve Customer Service By Reducing Wait Time For CSR	3	0	1	1																		1	66.67%
Integrate Real Time Assistance On The Online Portal	8	2	1	2																		3	62.50%
Improve Customer Satisfaction by Improving Information on the CSR	1	1	0	0																		0	100.00%
Test And Debug The Mobile App And CRM System	8	2	2	1																		3	62.50%
Finalize User Docmentation And Training Materials	5	0	1	2																		1	80.00%

5.2.3.2 Full



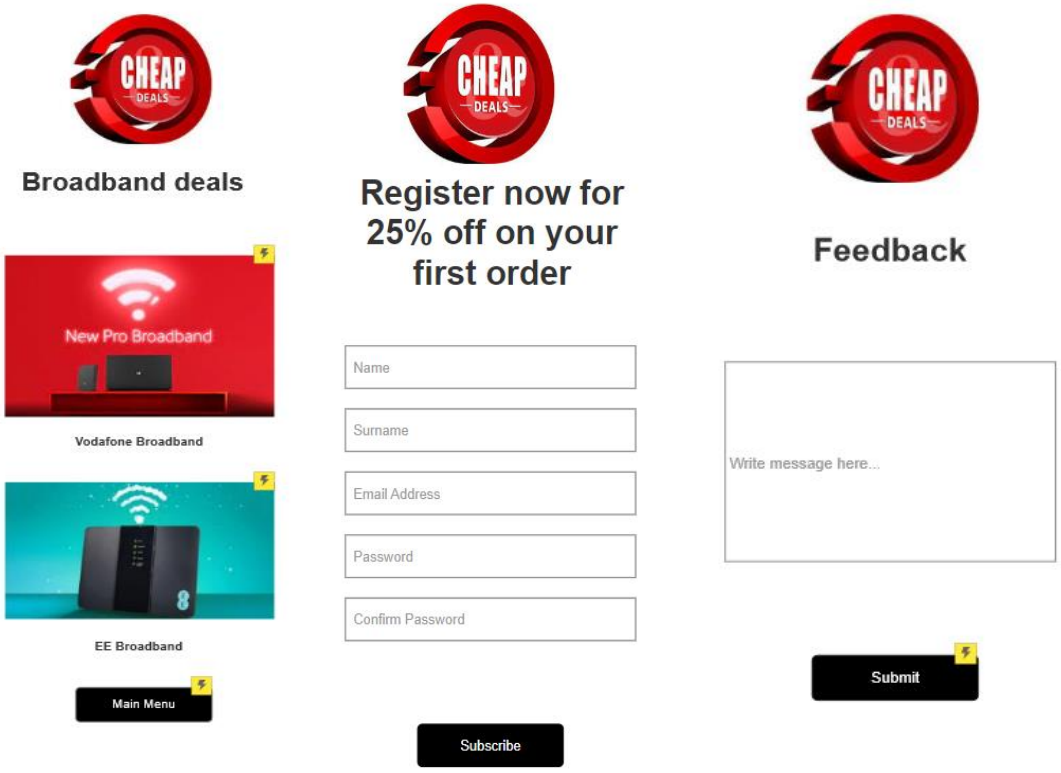
Burndown Chart

Iteration Number: 1
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User Stories		Days (Points Completed)																				Remaining	
Product Backlog Items	Beginning Balance	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Task Balance	Percentage Completed
Offer Special Deals By Profiling Customers	5	1	1	0	2	1																0	100.00%
Improve Customer Service By Reducing Wait Time For CSR	3	0	1	1	0	1																0	100.00%
Integrate Real Time Assistance On The Online Portal	8	2	1	2	2	1																0	100.00%
Improve Customer Satisfaction by Improving Information on the CSR	1	1	0	0	0	0																0	100.00%
Test And Debug The Mobile App And CRM System	8	2	2	1	1	2																0	100.00%
Finalize User Docmentation And Training Materials	5	0	1	2	1	1																0	100.00%

In the screenshots above, we have showcased the burndown charts that we utilised in our third sprint to be able to see the completion of our sprint in contrast to our projection.

5.2.4 Prototype



In the screenshots above, we have showcased the page with the broadband deals, registration form, as well as the feedback form page.

5.2.5 Sprint Daily Meeting

Duration of Sprint: 5 Days

Sprint Start: 17th March

Sprint End: 21st March

5.2.6 Sprint Review

The team worked on a number of features during this sprint, including order tracking, order filtering, order sorting, task filtering, task progress tracking, reporting page sales, admin dashboard, and task search capabilities.

Oviyan worked on order tracking and workflow while Tejesh focused on the integration of the admin dashboard and payment gateway. As Dimitrios worked on the task search and product search features, Anayat worked on the sales report page, order tracking, and product search features. Working with clients to meet their demands is part of Marini's responsibilities.

Throughout the sprint, the team reported no barriers or delays, indicating a seamless process. Throughout the sprint, each team member has duties they want to do, and the majority of them want to complete the work they started at the beginning of the sprint.

The sprint was successful overall, as the team finished most of the tasks that were given to it. They provided what was required of them and met their objectives.

5.2.7 Sprint Retrospective

Execute the essential duties of an e-commerce website for the sprint.

Tejesh, Oviyan, Anayat, Dimitris, and Marini were present.

What was successful:

During the sprint, the team was able to finish the tasks that were given to them.

There were no blockers during the Sprint.

Each team member may concentrate on the responsibilities they have been given, allowing the development process to go forward.

During the daily scrum sessions, the team efficiently communicated and exchanged updates on their progress. What might be done better:

More communication between team members would be beneficial, especially for projects that call for cooperation.

To guarantee that all tasks are finished by the deadline, the team may think about creating more precise and quantifiable goals for the sprint.

To make sure that the most crucial activities are completed first, it would be beneficial to prioritise jobs more systematically.

Overall, it appears like the team is working well on its duties and made strong progress throughout the sprint. The team may continue to enhance its procedures and increase productivity in next sprints by examining areas for improvement.

Individual Report – Reflection and Evaluation (Section 2)

6.1 Discussion of the course and what was learnt

Scrum was a new and interesting experience. Up to this date, I was only used to the waterfall methodology. The initial transition was quite challenging because they both have different sets of methods and implementation. It made me feel very uncomfortable and felt like I was changing my root behaviour.

I did feel quite overwhelmed by the pace at which Scrum proceeds. The sprints were short and felt limiting compared to the development cycles which I was used to. The daily meetings felt very intense as well because we would have to explain what task we have accomplished. I started to enjoy the transparency and collaborative environment Scrum pushed out, I felt more in tune with my team, and it got me more invested in making each sprint cycle successful. It was an interesting and fulfilling role that comes from being part of a self-organizing team.

The most important thing I learned about scrum was flexibility and adaptability. The waterfall methodology used to have fixed requirements and timeline at the start of the project, however, this was not the case with Scrum, where the priorities kept being changed based on the outcomes of each sprint. I could make better decisions about where to focus my efforts even during uncertain circumstances, which allowed me to gain flexibility. I believe that these skills are very valuable as they will also be helpful in other aspects, other than Scrum.

Scrum improved my communication and collaborative skills. I noticed that I started to speak and share my ideas and concerns during the meeting. I also improved my listening skills, where I was able to listen to the feedback, I got from my team to implement into my work. This helped me understand the environment Scrum creates, and that an effective communicative environment can go a long way to contribute to the success of any project.

Looking back on this interesting Scrum project, I can get a sense of how Scrum helped me develop personally and professionally. It helped me get comfortable working with change and uncertainty when working as a team while being transparent. While Scrum has its challenges, I believe it carries more positives and it completely changes the whole perspective of working as a team.

6.2 Discussion of teamwork and assessment of team members

I thought that forming a group with a balanced skill set would be quite difficult, however, that was the easiest part for our group. We all had already worked as a group in a previous waterfall-modelled project before, therefore we knew how each other performs and their skill set.

At first, we all had to sit down and understand what Scrum is and what we had to do during this project. During our first meetup, we had only come up with the product vision and turned the requirements into user stories. During this time, we understood that it will be next to impossible to complete the project within the allotted timeframe, so we had to decide that some tasks should be done individually. We initially disagreed, because if we had done this it would feel just like the waterfall methodology, where we would just compile everything together at the end.

However, I felt that this was a much better approach because we finished our task at a faster rate, and other group members were also more indulged with what results from others produced, this way we had way more ideas and questions generated. During this phase of the product backlog Marini, Dimitrios, and Oviyan had done the requirements and epic stories, and acceptance criteria, Anayat and I worked on the Moscow Prioritization and MVP.

During the Planning Poker, we all met up to discuss what we were going to do because we had never done anything close to this. After we understood the concept, we finally started. Initially, we had many arguments about the point system, but we would then explain our point of view. This part helps the bonding and trust process of the group. We were able to successfully improve our collaborative and communication skills.

We then elected Marini as the Scrum Master, and the rest of us were the developers. Marini was the Scrum Master in name, and we all did the Scrum Master task, to gain an understanding of the kind of task a Scrum Master carries out during Scrum.

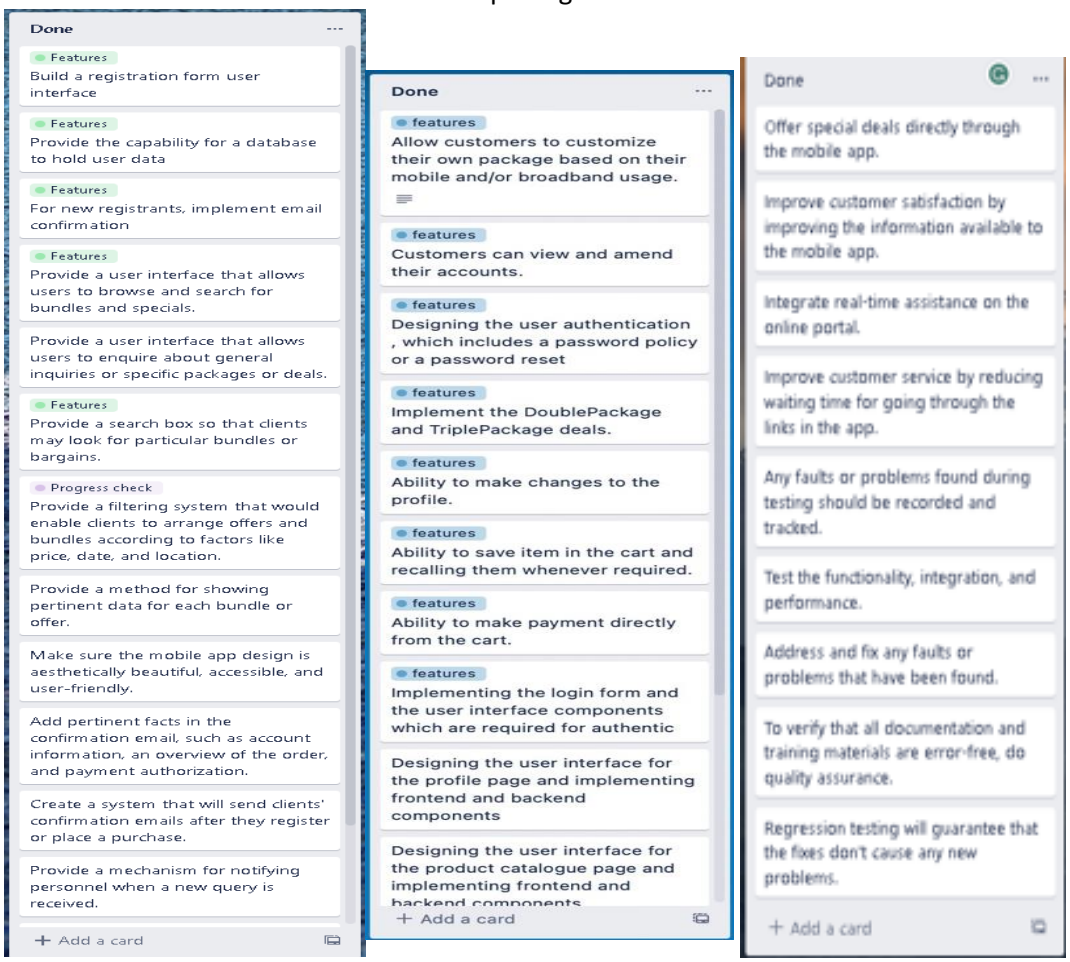
During each of the sprints, we would each come forward and choose what task we would be comfortable with. We want the divide and conquer route, where half of us would work on the report and the other half would work on the development of the prototype. If Marini, Oviyan, and I would be working on the report, then Anayat and Dimitrios would be working on the prototype and vice versa.

I believe we did a good job in developing a prototype using the Scrum Methodology, we hit all the checkpoints on our list and each member got to try out different aspects of the implementation of Scrum.

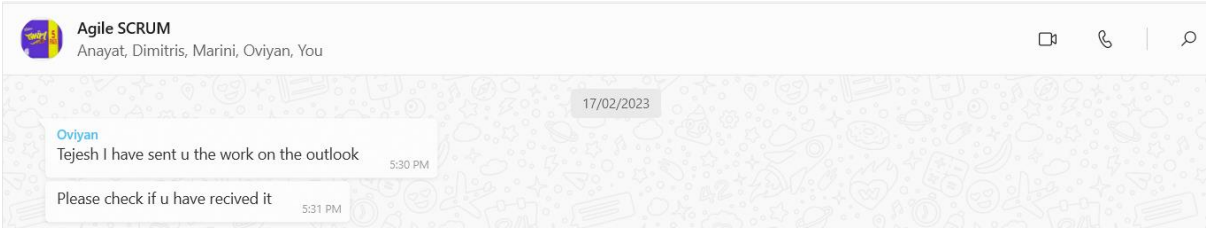
6.3 Sprint Retrospective

What went well?

During the Scrum process, all the tasks were achieved before the deadline, this only happened because we did a collective effort in completing the task that we had chosen for ourselves.

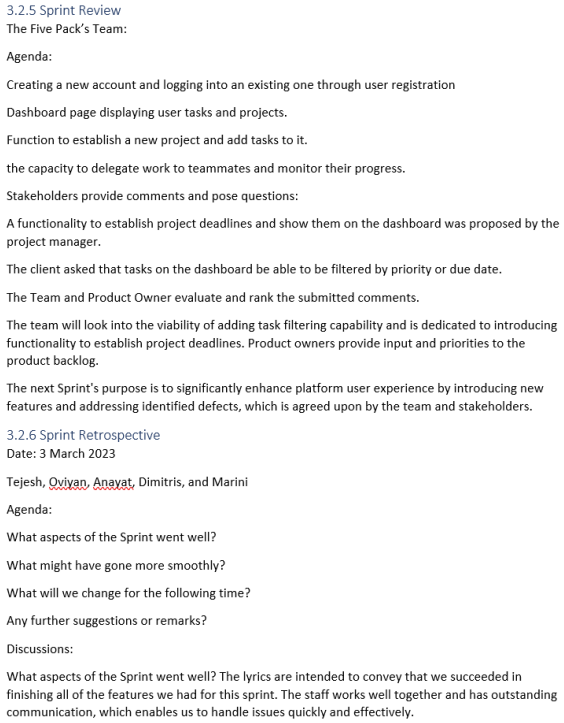


All the team members were flexible and understanding. If any of us had any problems whether it was a personal problem, like sickness or something we didn't understand we would just ask in our WhatsApp group chat. There was good open communication among the team members, and we all had some fun in the group chat, which made us all more open about what we did.



What went bad?

There was slight confusion when a task was handed to someone else. This is extremely bad because sometimes we couldn't understand what the other person had done, whereby we would just waste time understanding their part. Different sprint review and retrospective formats for different sprints



3.2.5 Sprint Review

The Five Pack's Team:

Agenda:

Creating a new account and logging into an existing one through user registration

Dashboard page displaying user tasks and projects.

Function to establish a new project and add tasks to it.

the capacity to delegate work to teammates and monitor their progress.

Stakeholders provide comments and pose questions:

A functionality to establish project deadlines and show them on the dashboard was proposed by the project manager.

The client asked that tasks on the dashboard be able to be filtered by priority or due date.

The Team and Product Owner evaluate and rank the submitted comments.

The team will look into the viability of adding task filtering capability and is dedicated to introducing functionality to establish project deadlines. Product owners provide input and priorities to the product backlog.

The next Sprint's purpose is to significantly enhance platform user experience by introducing new features and addressing identified defects, which is agreed upon by the team and stakeholders.

3.2.6 Sprint Retrospective

Date: 3 March 2023

Tejesh, Oviyan, Anayat, Dimitris, and Marini

Agenda:

What aspects of the Sprint went well?

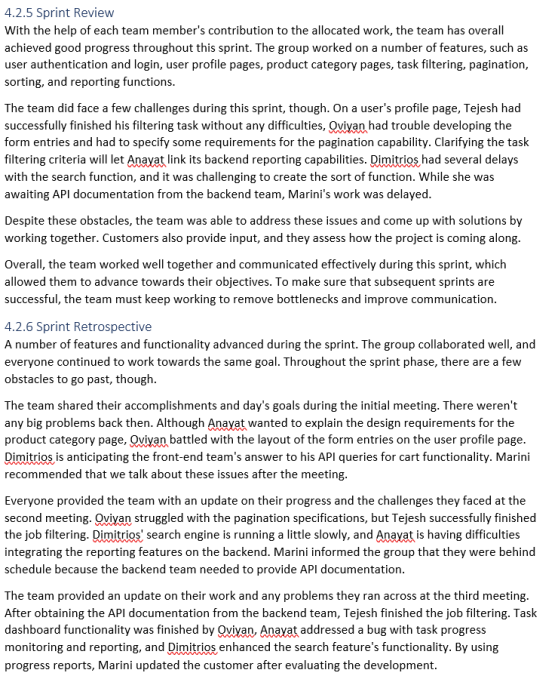
What might have gone more smoothly?

What will we change for the following time?

Any further suggestions or remarks?

Discussions:

What aspects of the Sprint went well? The lyrics are intended to convey that we succeeded in finishing all of the features we had for this sprint. The staff works well together and has outstanding communication, which enables us to handle issues quickly and effectively.



4.2.5 Sprint Review

With the help of each team member's contribution to the allocated work, the team has overall achieved good progress throughout this sprint. The group worked on a number of features, such as user authentication and login, user profile pages, product category pages, task filtering, pagination, sorting, and reporting functions.

The team did face a few challenges during this sprint, though. On a user's profile page, Tejesh had successfully finished his filtering task without any difficulties, Oviyan had trouble developing the form entries and had to specify some requirements for the pagination capability. Clarifying the task filtering criteria will let Anayat link its backend reporting capabilities. Dimitrios had several delays with the search function, and it was challenging to create the sort of function. While she was awaiting API documentation from the backend team, Marini's work was delayed.

Despite these obstacles, the team was able to address these issues and come up with solutions by working together. Customers also provide input, and they assess how the project is coming along.

Overall, the team worked well together and communicated effectively during this sprint, which allowed them to advance towards their objectives. To make sure that subsequent sprints are successful, the team must keep working to remove bottlenecks and improve communication.

4.2.6 Sprint Retrospective

A number of features and functionality advanced during the sprint. The group collaborated well, and everyone continued to work towards the same goal. Throughout the sprint phase, there are a few obstacles to go past, though.

The team shared their accomplishments and day's goals during the initial meeting. There weren't any big problems back then. Although Anayat wanted to explain the design requirements for the product category page, Oviyan battled with the layout of the form entries on the user profile page. Dimitrios is anticipating the front-end team's answer to his API queries for cart functionality. Marini recommended that we talk about these issues after the meeting.

Everyone provided the team with an update on their progress and the challenges they faced at the second meeting. Oviyan struggled with the pagination specifications, but Tejesh successfully finished the job filtering. Dimitrios' search engine is running a little slowly, and Anayat is having difficulties integrating the reporting features on the backend. Marini informed the group that they were behind schedule because the backend team needed to provide API documentation.

The team provided an update on their work and any problems they ran across at the third meeting. After obtaining the API documentation from the backend team, Tejesh finished the job filtering. Task dashboard functionality was finished by Oviyan, Anayat addressed a bug with task progress monitoring and reporting, and Dimitrios enhanced the search feature's functionality. By using progress reports, Marini updated the customer after evaluating the development.

We had off-timing because of everyone's availability. The meetings we held would sometimes go sideways, as we would get distracted by talking about other coursework. Some team members had issues connecting with the remote desktop to use the Axure.

What ideas doe we have?

We could have held more meetings to further include other group members in someone else's task to gain insight into it. This would have been very important because it would have eased the transition of the task shifting from one person to another.

What could we try?

We could use the computer at the university. We could have met up at the university more to use the available resources at the university.

My personal experience would be that this project was a very fun and iterative process, we had many ups and downs, however throughout this Scrum Process I learned that the end product is important, but it is the process that makes the end goal a fun journey.

Group/Team Name: Five Pack Group

Team Member Name	Student ID	Individual Overall Work Contribution (%)	Additional Notes On Task Description	Signature
Marini Tsako	001167815	20%	Scrum Master, User Stories, Scrum Board	Marini
Tejesh Ramesh Bawa	001173097	20%	Moscow Prioritization, MVP, Burndown charts	Tejesh
Dimitrios Antonis	001137051	20%	Prototype, poker	Dimitrios
Oviyan Sivanantham	001167908	20%	Acceptance Criteria, definition of done	Oviyan
Mohammad Anayatul Islam	001167908	20%	Backlog, Scrum Board, Use Case	Anayatul